

General: summarized references

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1. VALIDATION AND IMPACT ANALYSIS OF PROGNOSTIC CLINICAL PREDICTION RULES FOR LBP IS NEEDED: A SYSTEMATIC REVIEW

First Author: Robin Haskins

Summary:

The search retrieved over 10,000 records and yielded 35 studies describing 30 prognostic clinical prediction rules for nonsurgical LBP. Most rules were still early “development-only” tools; only three had validation evidence highlighted by the reviewers (including the Cassandra rule and Flynn manipulation rules). No study showed that applying a rule in practice improved patient outcomes or saved resources, so the review warns against treating most rules as ready for bedside use.

Date: 2015-07-01 Publication: Journal of Clinical Epidemiology Volume: 68 Issue: 7 Pages: 821-832 DOI: 10.1016/j.jclinepi.2015.02.003 <http://doi.org/10.1016/j.jclinepi.2015.02.003>

2. CAN SCREENING INSTRUMENTS ACCURATELY DETERMINE POOR OUTCOME RISK IN ADULTS WITH RECENT ONSET LBP? A SYSTEMATIC REVIEW AND META-ANALYSIS

First Author: Emma L. Karran

Summary:

Pooling studies on adults with LBP under three months old, the meta-analysis found screening tools more informative for predicting future disability than future pain. The Örebro questionnaire appeared among the stronger performers for some outcomes. The authors caution that misclassifying patients’ risk remains possible, so tools should not be treated as perfect predictors.

Date: 2017-01-19 Publication: BMC Medicine Volume: 15 Issue: 1 Pages: 13 DOI: 10.1186/s12916-016-0774-4 <http://doi.org/10.1186/s12916-016-0774-4>

3. A SURVEY ON DIALOGUE SYSTEMS: RECENT ADVANCES AND NEW FRONTIERS

First Author: Hongshen Chen

Summary:

Overview of deep-learning advances in dialogue systems, divided into task-oriented and non-task-oriented categories. Deep learning enables systems to learn feature representations and generation strategies from large data with minimal hand-crafting. While engineering-heavy approaches still dominate practical deployments, data-driven methods are increasingly feasible and promising.

Date: November 21, 2017 Publication: SIGKDD Explor. Newsl. Volume: 19 Issue: 2 Pages: 25–35
DOI: 10.1145/3166054.3166058 <http://doi.org/10.1145/3166054.3166058>

4. DIGITAL AND ONLINE SYMPTOM CHECKERS AND HEALTH ASSESSMENT/TRIAGE SERVICES FOR URGENT HEALTH PROBLEMS: SYSTEMATIC REVIEW

First Author: Duncan Chambers

Summary:

Systematic review of digital and online symptom checkers for people seeking help with urgent problems. Twenty-nine publications (27 studies) were included. Evidence on patient safety was weak; diagnostic accuracy varied across systems but was generally low; algorithm-based triage tended to be more risk-averse than health professionals. There was very limited evidence on compliance with online triage advice. Satisfaction was often high but largely in uncontrolled studies. Younger and more educated people were more likely to use these services. The authors conclude the digital service was introduced against substantial uncertainty about impact on important outcomes, with equity implications.

Date: 2019-08-01 Publication: BMJ Open Volume: 9 Issue: 8 Pages: e027743 DOI: 10.1136/bmjopen-2018-027743 <http://doi.org/10.1136/bmjopen-2018-027743>

5. THE EFFECTIVENESS OF DIGITAL HEALTH INTERVENTIONS IN THE MANAGEMENT OF MUSCULOSKELETAL CONDITIONS: SYSTEMATIC LITERATURE REVIEW

First Author: Stephanie Hewitt

Summary:

Nineteen trials met inclusion criteria for digital programs targeting musculoskeletal conditions. Nine trials reported statistically significant pain reductions, and ten of sixteen trials that measured function showed significant functional gains. Because study designs and outcomes differed so much, the authors could not combine results into one pooled effect.

Date: 2020-06-05 Publication: Journal of Medical Internet Research Volume: 22 Issue: 6 Pages: e15617 DOI: 10.2196/15617 <http://doi.org/10.2196/15617>

6. RISK-STRATIFIED AND STEPPED MODELS OF CARE FOR BACK PAIN AND OSTEOARTHRITIS: ARE WE HEADING TOWARDS A COMMON MODEL?

First Author: Alice Kongsted

Summary:

Compares evidence on stepped-care models for osteoarthritis with risk-stratified models for LBP- current data do not show one broad model type is clearly more effective than the other. Flags practical gaps: existing programs often ignore social context, other illnesses, and past treatment, and rarely link clinic care, self-management, and community supports in one system.

Date: September/October 2020 Publication: PAIN Reports Volume: 5 Issue: 5 Pages: e843 DOI: 10.1097/PR9.0000000000000843 <http://doi.org/10.1097/PR9.0000000000000843>

7. CHATBOTS: HISTORY, TECHNOLOGY, AND APPLICATIONS

First Author: Eleni Adamopoulou

Summary:

Literature review tracing chatbot evolution from early generative ideas to present-day systems. Two core implementation technologies are analyzed: pattern matching and machine learning. The authors present a general architectural design, highlight key pre-design considerations, survey industrial applications, and identify risks of chatbot use with suggested mitigations. The review concludes that chatbots still have significant room for improvement before becoming truly intelligent.

Date: 2020-12-15 Publication: Machine Learning with Applications Volume: 2 Pages: 100006
DOI: 10.1016/j.mlwa.2020.100006 <http://doi.org/10.1016/j.mlwa.2020.100006>

8. THE DIAGNOSTIC AND TRIAGE ACCURACY OF DIGITAL AND ONLINE SYMPTOM CHECKER TOOLS: A SYSTEMATIC REVIEW

First Author: William Wallace

Summary:

After 177 retrieved studies, 10 met inclusion (real or simulated patients; online/digital symptom checkers). Primary diagnostic accuracy was low across studies (range about 19–37.9%) and differed between checkers despite consistent symptom input. Triage accuracy was typically higher than diagnostic accuracy (range about 48.8–90.1%) but still variable. The authors conclude diagnostic and triage performance are variable and often weak, and that relying on symptom checkers could pose patient-safety risks unless large real-world studies and clearer regulation and implementation standards follow.

Date: 2022-08-17 Publication: npj Digital Medicine Volume: 5 Issue: 1 Pages: 118 DOI:
10.1038/s41746-022-00667-w <http://doi.org/10.1038/s41746-022-00667-w>

9. A SCOPING REVIEW ON THE USE AND USEFULNESS OF ONLINE SYMPTOM CHECKERS AND TRIAGE SYSTEMS: HOW TO PROCEED?

First Author: Anthony Pairon

Summary:

After screening 10,250 records, 28 publications were included. Users skew younger, female, more educated, and more technologically literate, with implications for equity and the digital divide. Triage algorithms remain risk-averse, which strains accuracy; newer algorithms show mixed success. Reported effects on demand, access, literacy, and surveillance vary widely. Patients and clinicians are generally positive, but heterogeneity across studies and systems limits transferability. The authors call for more comparable methods, setting-specific evaluations, regulation and standardization of assessments, and patient-centered design.

Date: 2023-01-06 Publication: Frontiers in Medicine Volume: 9 DOI:
10.3389/fmed.2022.1040926 <http://doi.org/10.3389/fmed.2022.1040926>

10. OVERVIEW OF CHATBOTS WITH SPECIAL EMPHASIS ON ARTIFICIAL INTELLIGENCE-ENABLED CHATGPT IN MEDICAL SCIENCE

First Author: Chiranjib Chakraborty

Summary:

Comprehensive review covering chatbot evolution, architecture, and medical applications, with a focus on ChatGPT. The article discusses ChatGPT's architecture and training, its use in medical diagnosis and treatment, ethical issues, and comparison with other NLP models. The authors conclude that large language models hold immense promise in healthcare but require further research, noting current limitations of ChatGPT-style tools in medical contexts.

Date: 2023-10-31 Publication: Frontiers in Artificial Intelligence Volume: 6 DOI: 10.3389/frai.2023.1237704 <http://doi.org/10.3389/frai.2023.1237704>

11. PROGNOSTIC MODELS FOR CHRONIC LBP OUTCOMES IN PRIMARY CARE ARE AT HIGH RISK OF BIAS AND LACK VALIDATION—HIGH-QUALITY STUDIES ARE NEEDED: A SYSTEMATIC REVIEW

First Author: Yanyan Fu

Summary:

From more than 5,500 screened records, only ten studies (34 prediction models) met the inclusion criteria. Reported model discrimination (AUC) ranged from 0.48 to 0.84, calibration was rarely reported, and risk-of-bias assessment raised serious concerns. The Örebro short questionnaire looked strongest among the candidates, but the reviewers do not recommend routine clinical adoption of these models until higher-quality development and external validation exist.

Date: 2024-04-29 Publication: Journal of Orthopaedic & Sports Physical Therapy Volume: 54 Issue: 5 Pages: 302-314 DOI: <https://www.jospt.org/doi/10.2519/jospt.2024.12081>

12. CRACKING THE CHRONIC PAIN CODE: A SCOPING REVIEW OF ARTIFICIAL INTELLIGENCE IN CHRONIC PAIN RESEARCH

First Author: Md Asif Khan

Summary:

60 studies met the reviewers' criteria. The included work most often reported AI success at diagnosing or classifying chronic pain, while relatively few studies focused on treatment, rehabilitation, or biological mechanisms of pain. Across papers, support vector machines, logistic regression, and random forests were the most common algorithm families. The take-home: AI publications cluster on diagnosis/classification, not on the kinds of mechanism-focused or treatment research the authors argue is needed next.

Date: 2024-05-01 Publication: Artificial Intelligence in Medicine Volume: 151 Pages: 102849 DOI: 10.1016/j.artmed.2024.102849 <http://doi.org/10.1016/j.artmed.2024.102849>

13. MACHINE LEARNING IN THE DIAGNOSIS, MANAGEMENT, AND CARE OF PATIENTS WITH LBP: A SCOPING REVIEW OF THE LITERATURE AND FUTURE DIRECTIONS

First Author: Andreas Seas

Summary:

The scoping review catalogued 223 machine-learning papers on LBP (1988–2023). More than half targeted detecting LBP from data; neural networks showed up in 106 articles. Common model inputs were history, demographics, and labs (about two-thirds of studies), and newer work often added imaging. Yet only eight studies reported external validation, only nine shared code, and the reviewers conclude that most papers fall short of best-practice ML reporting—making most of the literature hard to reproduce or trust for real-world use.

Date: 2025-01-01 Publication: The Spine Journal Volume: 25 Issue: 1 Pages: 18–31 DOI: 10.1016/j.spinee.2024.09.010 <http://doi.org/10.1016/j.spinee.2024.09.010>

14. PREVENTION AND MANAGEMENT OF DEGENERATIVE LUMBAR SPINE DISORDERS THROUGH ARTIFICIAL INTELLIGENCE-BASED DECISION SUPPORT SYSTEMS: A SYSTEMATIC REVIEW

First Author: Paolo Giaccone

Summary:

25 studies on AI decision-support systems applied to degenerative lumbar spine disorders and related LBP. The included papers span tasks such as defining clinical scores, automated assessment, and predicting who is eligible for certain pathways; the reviewers summarize that many studies report strong model metrics in their own datasets, while also highlighting heterogeneity in methods and endpoints across the field.

Date: 2025-02-07 Publication: BMC Musculoskeletal Disorders Volume: 26 Issue: 1 Pages: 126 DOI: 10.1186/s12891-025-08356-x <http://doi.org/10.1186/s12891-025-08356-x>

15. RIGHT DIAGNOSES BUT WRONG REASONING: CURRENT LARGE-LANGUAGE MODEL-BASED AGENTIC FRAMEWORKS HAVE FLAWED CLINICAL REASONING DESPITE HIGH DIAGNOSTIC ACCURACY

First Author: Prakashini Mruthyunjaya

Summary:

Preprint on LLM-based agentic frameworks for rheumatoid arthritis screening where workforce shortages delay care. The authors highlight that high diagnostic accuracy can coexist with flawed clinical reasoning—a barrier to safe deployment of mobile-accessible LLM screening—and discuss agentic designs (including a screening-oriented framework, SARA) in that context.

Date: 2025-07-09 Publication: SSRN (preprint) DOI: 10.2139/ssrn.5339074 <http://doi.org/10.2139/ssrn.5339074>

16. EVIDENCE FOR DIGITAL HEALTH TOOLS DESIGNED TO SUPPORT THE TRIAGE OF MUSCULOSKELETAL CONDITIONS IN PRIMARY, URGENT, AND EMERGENCY CARE SETTINGS: SCOPING REVIEW

First Author: Linda K. Truong

Summary:

34 studies (covering 37,509 patients) made the final map of digital triage tools for muscle and joint problems. Only a minority of tools were MSK-specific; most were generic symptom

checkers. Reported sensitivity and specificity varied widely (roughly 39–91% and 23–80% across tools), and reported accuracy ranged from 33% to 98%. The bottom line: few tools target MSK well, and many perform poorly on MSK populations, so digital output should support—but not replace—clinical judgment.

Date: 2026-01-14 Publication: Journal of Medical Internet Research Volume: 28 Issue: 1 Pages: e81578 DOI: 10.2196/81578 <http://doi.org/10.2196/81578>

OVERALL SYNTHESIS (FROM SUMMARIES, EMPHASIZING BOLDED CONCLUSIONS)

i) Main findings

Across systematic and scoping reviews, prognostic rules and models for low back pain (LBP) are numerous but rarely shown to change care or outcomes when used in practice: no included study demonstrated that applying a rule improved patient outcomes or saved resources, and recent reviews still do not recommend routine adoption until higher-quality development and external validation exist; discrimination is often modest and calibration under-reported. Screening tools appear more informative for future disability than for future pain, with instruments such as the Örebro questionnaire among stronger candidates, though misclassification remains a concern. Evidence on digital symptom checkers and online triage is weak on patient safety; diagnostic accuracy is generally low and variable (roughly 19–38% for top diagnosis), triage accuracy somewhat higher but still inconsistent (roughly 49–90%), algorithm-based triage tends to be more risk-averse than clinicians, and younger, more educated users are overrepresented, raising equity concerns. Reported effects on demand, access, literacy, and surveillance vary widely, and heterogeneity across studies and systems limits transferability. Despite that, digital musculoskeletal interventions often report statistically significant pain and/or function improvements in trials, though effects cannot be pooled simply. Stepped and risk-stratified care programs often fail to integrate social context, multimorbidity, prior treatment, and links between clinic, self-management, and community.

Artificial intelligence work in chronic pain and LBP clusters on diagnosis, detection, and classification rather than treatment, mechanisms, or rehabilitation; machine learning studies rarely report external validation or share code, limiting reproducibility and trust. Large language models can show high diagnostic accuracy alongside flawed clinical reasoning, and few tools target musculoskeletal (MSK) triage well; many perform poorly on MSK populations, so output should support but not replace clinical judgment. Broader dialogue-system and chatbot literature provides important context: deep learning has made data-driven dialogue increasingly feasible, but engineering-heavy approaches still dominate practical deployments; chatbot architectures rest on pattern matching and machine learning as core technologies, with significant room for improvement before truly intelligent behavior; and while LLMs like ChatGPT hold immense promise in healthcare, they require further research and current limitations in medical contexts are acknowledged.

ii) Gaps and opportunities

There is a clear gap between development of prediction tools and prospective evidence that their use improves outcomes, efficiency, or safety in real pathways. Opportunities include: rigorous external validation and calibration reporting; trials that embed tools in workflow and measure harms, equity, and resource use; MSK-specific triage and education tools with transparent performance in MSK cohorts; regulatory clarity and standardized, comparable evaluation methods for digital triage, including setting-specific evaluations, regulation and standardization of assessments, and patient-centered design; research that shifts AI from classification toward

treatment support, prognosis under intervention, and mechanistic insight; and reproducible ML (shared code, preregistered benchmarks) plus explicit testing of whether apparent LLM accuracy reflects safe reasoning for deployment decisions. The dialogue-systems literature highlights that transfer learning between datasets and incorporation of external knowledge remain underexplored opportunities for health applications, and that bridging the gap between data-driven dialogue methods and clinical deployment requires attention to both technical maturity and domain-specific risks.